CareerPilot: Stage 2 Implementation Report

Course	EECS 3311 Software Design, Fall 2026
Repository	https://github.com/Parsa-Meshkini/CareerPilot
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Generated	2026-11-13 by tools/build_stage2_report.py

1. Project overview and implementation status

CareerPilot is a desktop and command-line assistant for students looking for co-op, internship and entry-level jobs. It keeps a profile of facts the student has verified, imports postings (browser extension, paste, file), scores them against the student's goal, tracks each application through its lifecycle, reads job-related emails, and prepares tailored documents and interviews. An AI agent plans and runs multi-step goals ("prepare my application for the Acme posting") and asks before anything that changes data.

AI/LLM integration. One facade (`LlmClient`) calls Gemini (default, free tier) or any OpenAI-compatible endpoint through the `LlmProvider` adapter. Prompts are versioned JSON files in `careerpilot-core/src/main/resources/prompts`, answers are validated against a schema, token budgets are enforced per day and per run, and every feature has a deterministic fallback when the model is unavailable. The agent uses planning, tool use, memory and multi-step execution with approval.

Working functionality. All 19 Stage 1 features are implemented and reachable from the CLI and the JavaFX GUI; F04 also needs the Chrome extension and the REST server. The Java modules are `careerpilot-core`, `careerpilot-cli`, `careerpilot-gui`, `careerpilot-server` plus `extension/`. Automated tests: core 124, server 3, CLI 7, GUI 1, extension (node) tests.

Honest limitations.

- Gmail sign-in (F14) is verified only against a local stand-in for Google's endpoints. It has not been run with a real Google account, which needs the student's own OAuth client.
- On Nov 3 the default Gemini model answered HTTP 503 ("high demand") for every attempt, so the live end-to-end check was run through the OpenAI-compatible provider instead (Section 6.2). The Gemini provider itself was verified live in an earlier session.
- LinkedIn extraction exists but is switched off by default (FEATURE_LINKEDIN_EXTRACTION=false).
- The full deterministic test suite and KUMA agent validation belong to Stage 3.

2. Feature to implementation mapping

Every Stage 1 feature (F01-F19) is accounted for; there are no new features beyond Stage 1 and none were dropped. The feature table below is machine-checked against the source: each class listed exists and each method listed appears in those classes. Video links are in the demonstration index (Section 7).

F01 Candidate profile and verified facts

Field	Entry
Feature ID and name	F01 - Candidate profile and verified facts
Stage 1 reference	F01
Status or change	Modified
Type	Deterministic
Implemented classes	`CandidateService`, `CandidateProfile`, `ProfileFact`, `FactType`, `FactSource`, `FactValidator`
Key methods	`updateProfile`, `addFact`, `verifyFacts`, `rejectFacts`, `completenessHints`
Design patterns	Repository (Spring Data)
Where to use it	GUI Profile page; CLI `profile show/set/fact-*`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. The view calls CandidateService, which validates the fact JSON with FactValidator against its FactType, stores it as an unverified ProfileFact, and only facts the student verifies are returned to the AI context. Stage 1 had one subclass per fact kind; one entity plus an enum with validated JSON replaced them.

F02 Resume import and AI fact extraction

Field	Entry
Feature ID and name	F02 - Resume import and AI fact extraction
Stage 1 reference	F02
Status or change	Implemented
Type	Hybrid
Implemented classes	`CandidateService`, `ResumeParser`, `LlmClient`, `StorageService`, `TaskQueue`, `UploadedResume`
Key methods	`importResume`, `parseResume`, `extractText`, `extractFacts`, `generateStructured`
Design patterns	Facade (LlmClient); Adapter (LlmProvider)
Where to use it	GUI Profile > Import resume; CLI `profile resume FILE`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. importResume stores the file through StorageService and queues parseResume on TaskQueue. ResumeParser extracts text (PDFBox, POI or plain) and asks LlmClient for facts, which are saved unverified for the student to review.

F03 Career goal and strategy configuration

Field	Entry
Feature ID and name	F03 - Career goal and strategy configuration
Stage 1 reference	F03

Field	Entry
Status or change	Implemented
Type	Deterministic
Implemented classes	`CandidateService`, `CareerGoal`, `GoalType`, `StrategyWeights`, `MatchingService`
Key methods	`updateCareerGoal`, `selectStrategy`
Design patterns	Strategy
Where to use it	GUI Profile > Goal; CLI `profile goal-set/goal-show`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. The goal type chosen by the student selects which MatchScoringStrategy MatchingService uses and seeds the StrategyWeights that F19 later adjusts.

F04 Browser-extension batch job import

Field	Entry
Feature ID and name	F04 - Browser-extension batch job import
Stage 1 reference	F04
Status or change	Implemented
Type	Deterministic
Implemented classes	`ApiController`, `TokenAuthInterceptor`, `JobImportService`, `JobSourceRegistry`, `YorkBrowserImportAdapter`, `ImportBatch`, `TaskQueue`
Key methods	`createBatch`, `processBatch`, `parse`
Design patterns	Adapter; Observer (JobsImported)
Where to use it	Chrome extension (extension/content, popup, api); REST `/api/import`; CLI `jobs import-york`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. The content script reads the postings on the page, the service worker posts them to ApiController with the student's token, JobImportService creates an ImportBatch and processes it in the background through the York adapter, then publishes JobsImported which triggers F09.

F05 Manual job import (paste or file)

Field	Entry
Feature ID and name	F05 - Manual job import (paste or file)
Stage 1 reference	F05
Status or change	Implemented
Type	Hybrid
Implemented classes	`JobImportService`, `ManualJobImportAdapter`, `MockJobSourceAdapter`, `LlmClient`
Key methods	`importManual`, `importRecord`, `parse`, `structure_paste`

Field	Entry
Design patterns	Adapter; Facade
Where to use it	GUI Jobs > Import; CLI `jobs import-paste/import-file`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6I2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. Pasted text goes to LlmClient (job.structure_paste prompt) to extract title, company and description; CSV or JSON files are read by the adapters directly. Both end in JobImportService.importRecord.

F06 Job normalization and duplicate detection

Field	Entry
Feature ID and name	F06 - Job normalization and duplicate detection
Stage 1 reference	F06
Status or change	Implemented
Type	Deterministic
Implemented classes	`JobNormalizer`, `DuplicateDetector`, `JobPosting`, `JobPostingRepository`, `ImportBatch`
Key methods	`normalize`, `findDuplicate`
Design patterns	Repository
Where to use it	Runs inside every import
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6I2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. importRecord calls JobNormalizer.normalize (clean text, deadline, work model, fingerprint) then DuplicateDetector.findDuplicate; a duplicate is merged and counted in the batch instead of saved again.

F07 AI job description analysis

Field	Entry
Feature ID and name	F07 - AI job description analysis
Stage 1 reference	F07
Status or change	Implemented
Type	AI
Implemented classes	`JobAnalysisService`, `JobAnalysis`, `LlmClient`, `LlmProvider`, `PromptRegistry`, `ResponseValidator`, `UsageTracker`
Key methods	`analyzeJob`, `generateStructured`, `validate`, `fallbackKeywordAnalysis`
Design patterns	Facade; Adapter
Where to use it	GUI Jobs detail; CLI `jobs analyze`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6I2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. analyzeJob sends the posting through LlmClient, which loads the versioned prompt, checks the token budget, calls the provider, validates the JSON against the schema, retries transient and schema failures with backoff, and records usage. If the model is down a keyword analysis is used and the result is marked as such.

F08 Candidate-job matching and skill-gap analysis

Field	Entry
Feature ID and name	F08 - Candidate-job matching and skill-gap analysis
Stage 1 reference	F08
Status or change	Implemented
Type	Hybrid
Implemented classes	`MatchingService`, `ContextBuilder`, `MatchScoringStrategy`, `CoopInternshipStrategy`, `EntryLevelStrategy`, `CareerSwitchStrategy`, `JobMatch`
Key methods	`matchJob`, `selectStrategy`, `score`, `rankJobs`, `buildCandidateContext`
Design patterns	Strategy
Where to use it	GUI Jobs (bearing ring, detail); CLI `jobs match/score/show`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. ContextBuilder gathers verified facts and the goal; MatchingService picks the strategy for the goal type and scores deterministically (quick or deep with the analysis); the AI only writes the explanation and transferable-skill hints, never the score.

F09 Agent batch triage and shortlist

Field	Entry
Feature ID and name	F09 - Agent batch triage and shortlist
Stage 1 reference	F09
Status or change	Implemented
Type	AI (agent)
Implemented classes	`TriageTrigger`, `AgentController`, `Planner`, `PlaybookLibrary`, `Reasoner`, `ToolManager`, `AgentCommand`, `QuickScoreJobsCommand`, `SelectForDeepAnalysisCommand`, `DeepMatchJobsCommand`, `RankJobsCommand`, `MemoryManager`, `AgentRun`
Key methods	`handle`, `startTriage`, `runGoal`, `createPlan`, `execute`, `evaluate`, `remember`
Design patterns	Command; Observer; Facade
Where to use it	Automatic after an import; GUI Agent page; CLI `agent runs/show`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. TriageTrigger listens for JobsImported and starts a run through AgentController. The plan (hand-written playbook, or the LLM's plan when valid) is executed one AgentCommand at a time by ToolManager;

after each step Reasoner decides to continue, replan or stop, and every action is recorded in AgentActionRecord.

F10 Tailored resume with fabrication guard

Field	Entry
Feature ID and name	F10 - Tailored resume with fabrication guard
Stage 1 reference	F10
Status or change	Implemented
Type	Hybrid
Implemented classes	`DocumentGenerator`, `ResumeGenerator`, `FabricationGuard`, `DocumentRenderer`, `DocumentService`, `GeneratedDocument`
Key methods	`generate`, `check`, `renderDocx`, `renderPdf`, `regenerateFromEdit`
Design patterns	Template Method; Facade
Where to use it	GUI Tracker > Documents; CLI `docs resume`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. DocumentService calls the generator's final generate(): build context, ask the LLM, run FabricationGuard.check on every claim against verified facts, remove violating claims, then render. Subclasses only supply the prompt variables, claim extraction and block layout.

F11 Tailored cover letter

Field	Entry
Feature ID and name	F11 - Tailored cover letter
Stage 1 reference	F11
Status or change	Implemented
Type	Hybrid
Implemented classes	`DocumentGenerator`, `CoverLetterGenerator`, `FabricationGuard`, `DocumentRenderer`
Key methods	`generate`, `check`
Design patterns	Template Method; Facade
Where to use it	GUI Tracker > Documents; CLI `docs cover-letter`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. Same template as F10 with a different prompt, claim extractor and layout.

F12 Application package builder and ZIP export

Field	Entry
Feature ID and name	F12 - Application package builder and ZIP export
Stage 1 reference	F12
Status or change	Implemented
Type	Deterministic

Field	Entry
Implemented classes	`PackageAssembler`, `ApplicationPackage`, `BuildPackageCommand`, `DocumentService`, `StorageService`
Key methods	`buildPackage`, `download`
Design patterns	Command; Template Method (uses)
Where to use it	GUI Tracker > Build package; CLI `docs package/download`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. PackageAssembler generates any missing documents, adds the job description and a checklist, zips them through StorageService and records an ApplicationPackage; the agent reaches it through BuildPackageCommand.

F13 Application lifecycle tracking and undo

Field	Entry
Feature ID and name	F13 - Application lifecycle tracking and undo
Stage 1 reference	F13
Status or change	Implemented
Type	Deterministic
Implemented classes	`ApplicationService`, `Application`, `ApplicationState`, `SavedState`, `PreparedState`, `AppliedState`, `InterviewState`, `OfferState`, `RejectedState`, `StatusChange`
Key methods	`createApplication`, `transition`, `undoTransition`
Design patterns	State; Observer
Where to use it	GUI Tracker board; CLI `apps create/act/undo`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. transition asks the current ApplicationState object whether the action is legal and for the next state, saves a StatusChange (which makes undo possible), and publishes ApplicationStatusChanged.

F14 Email monitoring and application event detection

Field	Entry
Feature ID and name	F14 - Email monitoring and application event detection
Stage 1 reference	F14
Status or change	Implemented (Gmail verified only against a local test double)
Type	Hybrid
Implemented classes	`EmailSyncService`, `EmailProviderRegistry`, `GmailAdapter`, `GmailConnector`, `CredentialVault`, `EmlFileAdapter`, `MockEmailAdapter`, `EmailNormalizer`, `EmailEventDetector`, `HybridClassifier`, `RuleBasedClassifier`, `LlmEmailClassifier`, `ApplicationMatcher`, `DetectedEmailEvent`
Key methods	`syncAccount`, `importEmlFiles`, `detect`, `classify`, `match`, `confirmEvent`, `fetchNewMessages`, `connect`

Field	Entry
Design patterns	Adapter; Strategy
Where to use it	GUI Email page; CLI `email upload/sync-mock/connect-gmail/events/confirm`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. A provider adapter returns RawEmails; EmailNormalizer cleans them; EmailEventDetector runs the HybridClassifier (rules first, LLM when rules are unsure); ApplicationMatcher links the email to an application; a confident match is published as an event and the rest wait for the student to confirm.

F15 Event-driven updates, reminders and notifications

Field	Entry
Feature ID and name	F15 - Event-driven updates, reminders and notifications
Stage 1 reference	F15
Status or change	Implemented
Type	Deterministic
Implemented classes	`DomainEventBus`, `EventListener`, `ApplicationStatusUpdater`, `InterviewScheduler`, `MemoryRecorder`, `NotificationService`, `ReminderService`, `Notification`
Key methods	`publish`, `subscribe`, `checkDeadlines`, `add`, `markAllRead`
Design patterns	Observer; State; Command (undo)
Where to use it	GUI Dashboard feed; CLI `notifications`, `remind`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. Detected email events are published on DomainEventBus; ApplicationStatusUpdater moves the application, InterviewScheduler records the interview, MemoryRecorder saves what happened for the agent, and NotificationService stores a de-duplicated notification. ReminderService raises deadline notifications.

F16 Goal-driven agent workspace

Field	Entry
Feature ID and name	F16 - Goal-driven agent workspace
Stage 1 reference	F16
Status or change	Implemented
Type	AI (agent)
Implemented classes	`AgentController`, `Planner`, `PlaybookLibrary`, `Reasoner`, `ToolManager`, `ApprovalPolicy`, `AgentCommand`, `UpdateApplicationStatusCommand`, `MemoryManager`, `ContextBuilder`, `AgentRun`, `AgentActionRecord`
Key methods	`runGoal`, `resumeRun`, `createPlan`, `requiresApproval`, `execute`, `undo`, `recall`
Design patterns	Command; Facade
Where to use it	GUI Agent page; CLI `agent run/approve/reject/undo/memory`

Field	Entry
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. A goal in plain words becomes a typed run; steps with side effects stop at ApprovalPolicy until the student approves, rejects (the agent replans and remembers the preference) or undoes; MemoryManager recalls preferences into later plans.

F17 Interview preparation plan

Field	Entry
Feature ID and name	F17 - Interview preparation plan
Stage 1 reference	F17
Status or change	Implemented
Type	AI
Implemented classes	`InterviewPrepService`, `InterviewPrepPlan`, `InterviewPrepTrigger`, `ContextBuilder`, `FabricationGuard`, `LlmClient`, `PrepareInterviewCommand`
Key methods	`preparePlan`, `latest`, `check`
Design patterns	Facade; Observer (auto-trigger)
Where to use it	GUI Interview page; CLI `interview prepare/show`; agent tool prepare_interview
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. InterviewPrepTrigger reacts to an interview status change; InterviewPrepService builds the context, asks the LLM for questions and STAR stories, and FabricationGuard rejects stories that use facts the student has not verified; a deterministic plan is used when the model fails.

F18 AI mock interview and feedback

Field	Entry
Feature ID and name	F18 - AI mock interview and feedback
Stage 1 reference	F18
Status or change	Implemented
Type	AI
Implemented classes	`MockInterviewService`, `MockInterviewSession`, `LlmClient`, `MemoryManager`
Key methods	`startSession`, `submitAnswer`, `finishSession`, `abandon`
Design patterns	Facade
Where to use it	GUI Interview page; CLI `interview mock/answer/finish`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. startSession asks for a first question; each submitAnswer is scored against a rubric by the LLM and the next question depends on the weak areas so far; finishSession writes a summary and stores

feedback in agent memory. Sessions can be resumed.

F19 Outcome analytics and adaptive strategy

Field	Entry
Feature ID and name	F19 - Outcome analytics and adaptive strategy
Stage 1 reference	F19
Status or change	Implemented
Type	Hybrid
Implemented classes	`AnalyticsService`, `OutcomeAnalyzer`, `StrategyAdvisor`, `StrategyAdjustment`, `StrategyWeights`, `MemoryManager`
Key methods	`getInsights`, `computeStats`, `proposeAdjustments`, `accept`, `reject`, `revert`
Design patterns	Strategy; Observer (MemoryRecorder feeds data)
Where to use it	GUI Insights page; CLI `analytics`
Video link/timestamp	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

How the classes collaborate. OutcomeAnalyzer computes response and interview rates by score band, source and role; StrategyAdvisor proposes bounded weight changes only with enough evidence; the LLM writes the explanation; nothing changes until the student accepts, and accepted changes can be reverted.

3. Implemented design patterns

Seven distinct patterns are implemented. Class diagrams generated from the code (UMLet sources in `docs/stage2/uml/src`) follow the table.

3.1 Facade

- **Problem addressed.** Callers should not know about prompts, schemas, budgets, retries or vendors, nor about the plan-act-observe loop.
- **Participants.** LlmClient (facade) over PromptRegistry, ResponseValidator, TokenBudget, UsageTracker, LlmProviderFactory; AgentController (facade) over Planner, ToolManager, Reasoner, ApprovalPolicy, MemoryManager.
- **Source files and methods.** llm/LlmClient.java generateStructured; agent/AgentController.java runGoal, resumeRun
- **How it follows the pattern and why it helps.** Every AI feature calls one method and gets validated JSON or a typed error; the agent is started with one call. Matches Stage 1.

3.2 Adapter

- **Problem addressed.** Job sources, mailboxes and LLM vendors all differ in format and protocol but the rest of the system needs one shape.
- **Participants.** JobSourceAdapter with ManualJobImportAdapter, YorkBrowserImportAdapter, MockJobSourceAdapter; EmailProviderAdapter with GmailAdapter, EmlFileAdapter, MockEmailAdapter;

LlmProvider with GeminiProvider, OpenAiCompatibleProvider, MockLlmProvider.
- **Source files and methods.** jobs/JobSourceAdapter.java parse; integrations/EmailProviderAdapter.java fetchNewMessages; llm/LlmProvider.java complete
- **How it follows the pattern and why it helps.** New sources are added by writing one class; GmailAdapter was added in Period 3 without touching the pipeline after the fetch. Differs from Stage 1 only in names (OpenAIProvider became OpenAiCompatibleProvider so Groq and OpenRouter work too).

3.3 Strategy

- **Problem addressed.** Scoring and classification rules differ by situation and must be swappable at run time.
- **Participants.** MatchScoringStrategy with CoopInternshipStrategy, EntryLevelStrategy, CareerSwitchStrategy chosen by MatchingService.selectStrategy; EmailClassificationStrategy with RuleBasedClassifier, LlmEmailClassifier, HybridClassifier held by EmailEventDetector.setStrategy.
- **Source files and methods.** jobs/MatchingService.java selectStrategy; integrations/EmailEventDetector.java detect, setStrategy
- **How it follows the pattern and why it helps.** The student's goal changes how jobs are scored without any if/else in the service; the email detector can run rules only, LLM only, or hybrid. Matches Stage 1.

3.4 Command

- **Problem addressed.** Every agent tool must be runnable, loggable, approvable and, where possible, undoable in the same way.
- **Participants.** AgentCommand (command) with MatchJobCommand, RankJobsCommand, UpdateApplicationStatusCommand, BuildPackageCommand, PrepareInterviewCommand and others; ToolManager (invoker); AgentActionRecord (history for undo).
- **Source files and methods.** agent/AgentCommand.java execute, undo; agent/ToolManager.java execute, undo
- **How it follows the pattern and why it helps.** The planner only names tools; ToolManager runs them uniformly, records each action and can undo reversible ones. Matches Stage 1.

3.5 State

- **Problem addressed.** Which actions are legal depends on where the application is in its lifecycle, and the rules must not be scattered across services.
- **Participants.** ApplicationState (abstract state) with SavedState, PreparedState, AppliedState, InterviewState, OfferState, AcceptedState, RejectedState, DeclinedState, WithdrawnState; Application (context).
- **Source files and methods.** applications/ApplicationState.java; applications/ApplicationService.java transition, undoTransition
- **How it follows the pattern and why it helps.** An illegal transition raises InvalidTransitionException from the state object itself. Matches Stage 1.

3.6 Observer

- **Problem addressed.** Importing, emailing and status changes must trigger reminders, memory, triage and interview prep without the publisher knowing the receivers.
- **Participants.** DomainEventBus (subject), EventListener (observer), listeners TriageTrigger, ApplicationStatusUpdater, InterviewScheduler, MemoryRecorder, InterviewPrepTrigger; events in Events.
- **Source files and methods.** events/DomainEventBus.java publish, subscribe; notifications/*.java handle

- **How it follows the pattern and why it helps.** A listener failure is collected and logged and never breaks the publisher or other listeners. Matches Stage 1.

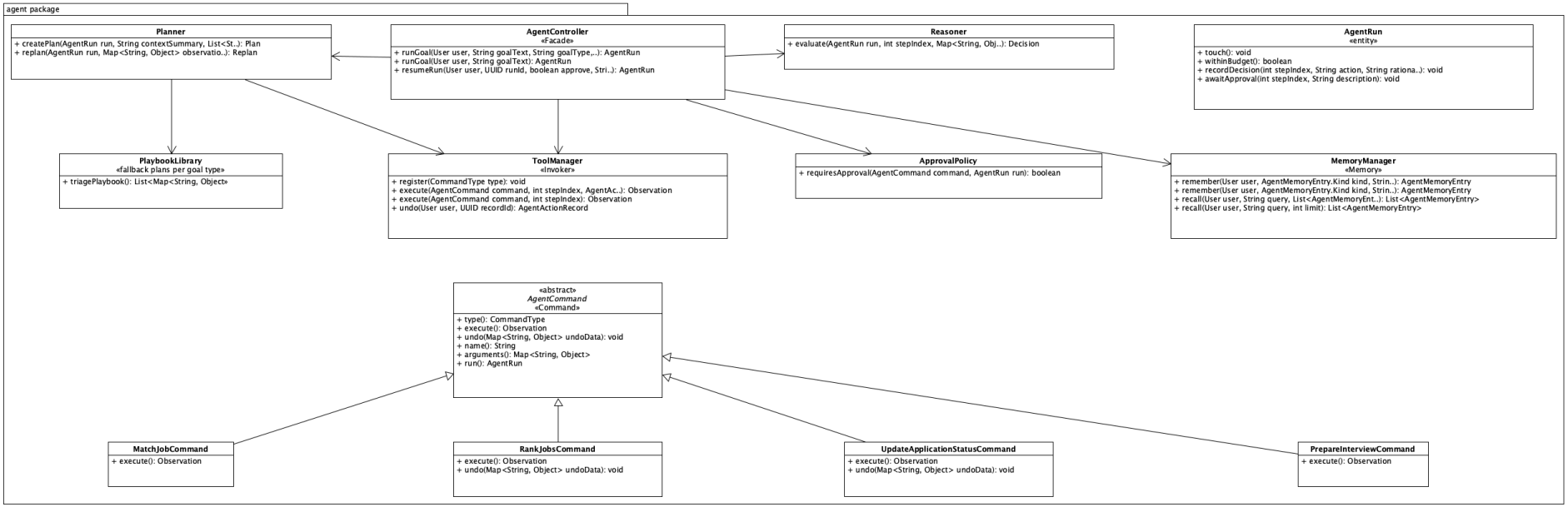
3.7 Template Method

- **Problem addressed.** Resume and cover letter generation share the same safety-critical steps and must not be able to skip the fabrication check.
- **Participants.** DocumentGenerator (final generate defines the steps; abstract promptKey, buildPromptVariables, claims, removeClaims, render) with ResumeGenerator and CoverLetterGenerator.
- **Source files and methods.** documents/DocumentGenerator.java generate; ResumeGenerator.java; CoverLetterGenerator.java
- **How it follows the pattern and why it helps.** generate is final, so FabricationGuard.check always runs. Matches Stage 1.

3.8 Updated UML class diagrams

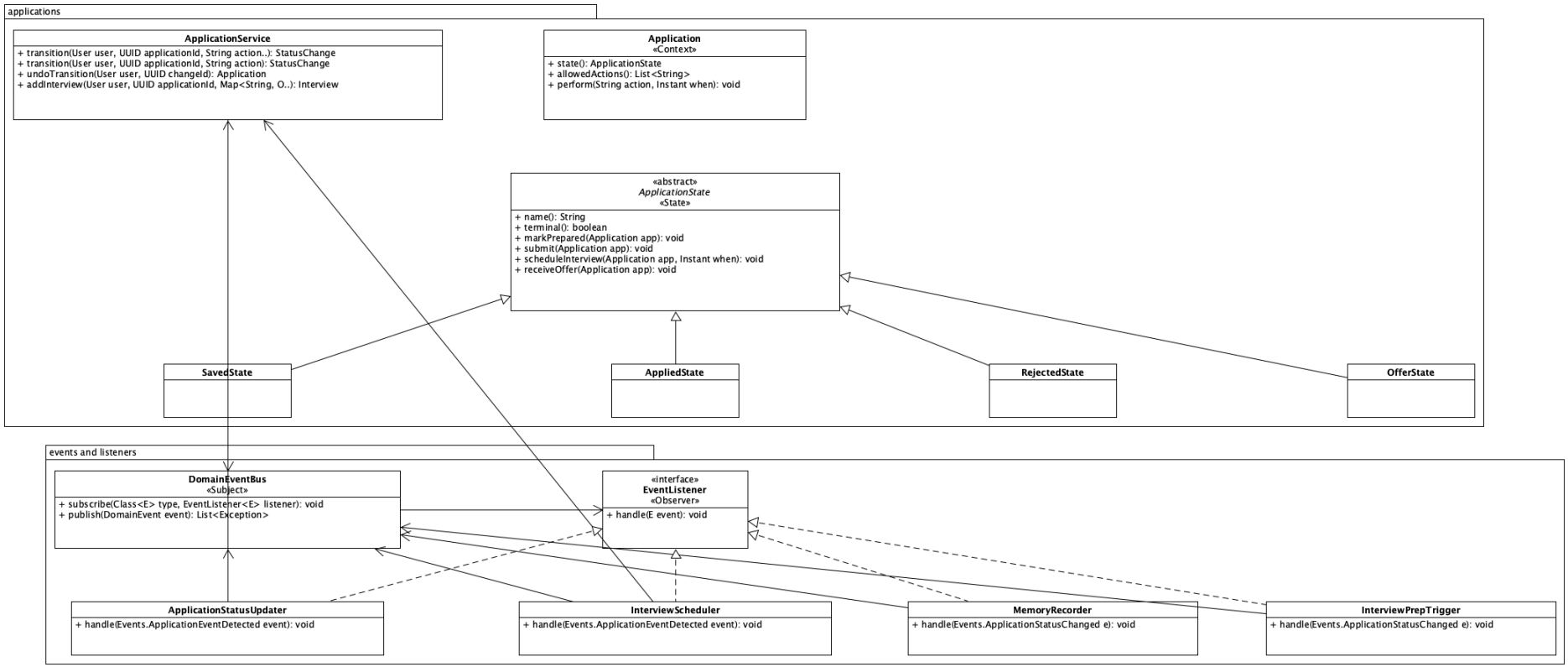
The Stage 1 class diagrams (`docs/uml`) described the design; the diagrams below are generated from the implemented Java classes by `tools/gen_stage2_uml.py`, so every box, signature and arrow exists in the code. The Stage 1 use-case and sequence diagrams are still accurate in behaviour; the only visible differences are the class names listed in Section 4 (for example LLMClient is LlmClient, GmailAdapter and CredentialVault are in the integrations package) and that UI classes are JavaFX views instead of web pages.

CD-S2-1 Agent: Command and Facade (F09, F16)
AgentController is the facade over the plan-act-observe loop. Each tool is an AgentCommand that ToolManager runs and can undo.



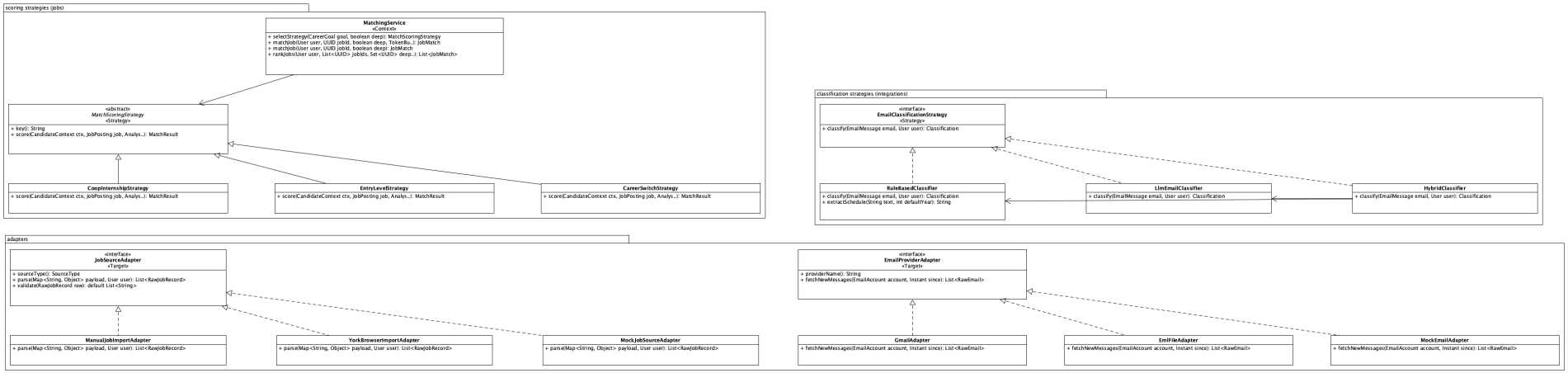
cd-s2-01_agent_command_facade

CD-S2-2 Application lifecycle State and Observer event bus (F13, F15)
ApplicationState subclasses own the legal transitions. DomainEventBus delivers events to EventListener implementations without the publisher knowing them.



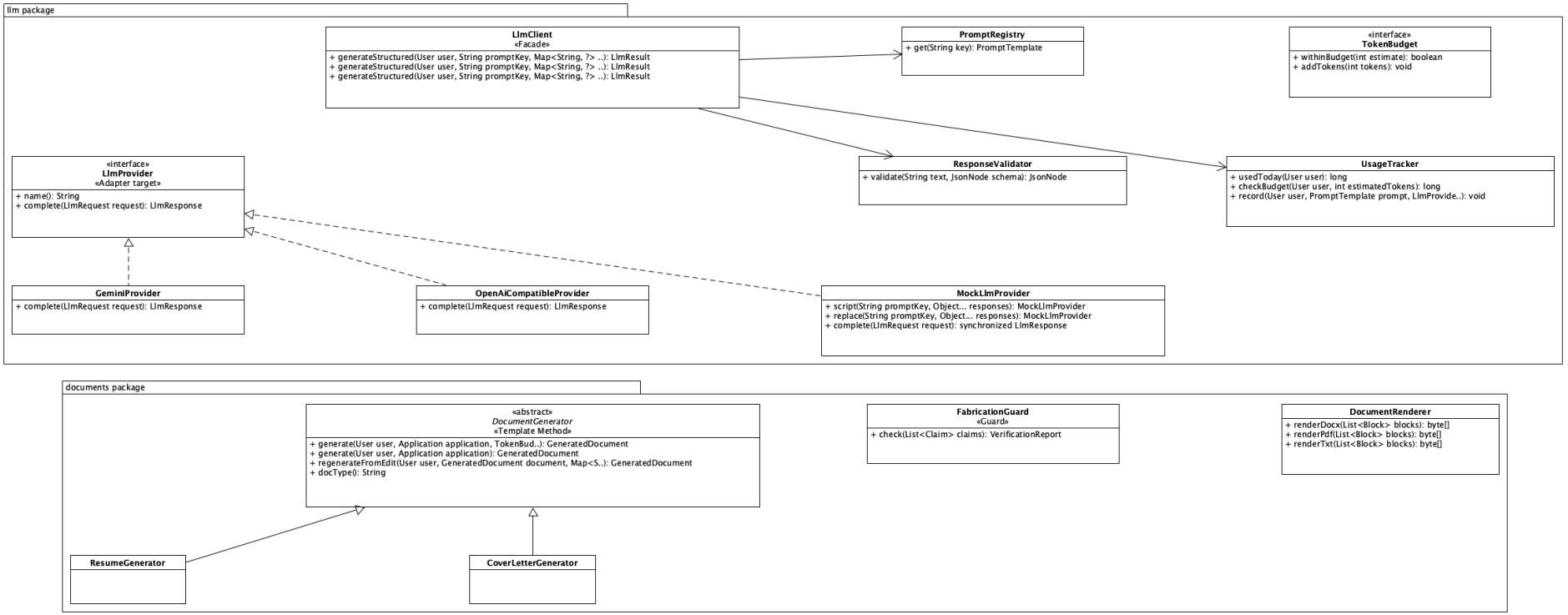
cd-s2-02_application_state_observer

CD-S2-3 Strategy and Adapter (F03, F04, F05, F08, F14)
Match scoring and email classification are interchangeable strategies; job sources, mail providers and LLM vendors sit behind adapter interfaces.



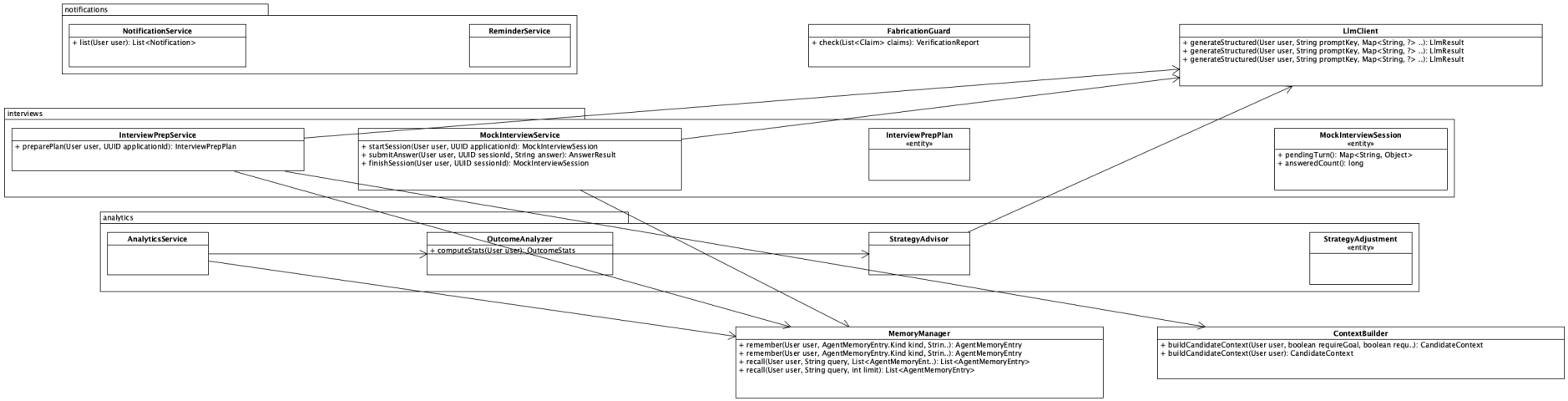
cd-s2-03_strategy_adapter

CD-S2-4 LLM Facade, provider Adapter and document Template Method (F02, F07, F10, F11)
LlmClient hides prompts, validation, budgets and vendors behind one call. DocumentGenerator fixes the generation steps; subclasses fill in the parts that differ.



cd-s2-04_llm_facade_templates

CD-S2-5 Period 3 features: notifications, interview preparation, mock interview, analytics (F15, F17, F18, F19)
These services share ContextBuilder, FabricationGuard, LlmClient and MemoryManager with the rest of the system, and are triggered by the domain event bus.



cd-s2-05_interviews_analytics_notifications

4. Changes and rationale

Original design	Implemented change	Reason	Impact
Python/Django REST backend, Celery and Redis	Java 17 and Spring Boot 3.3, in-process TaskQueue	The course requires Java for Stage 2; a single process removes the need to run a broker for a desktop tool.	All classes keep their Stage 1 names in Java camelCase; background work is a TaskQueue instead of Celery tasks.
React single-page app	JavaFX GUI with AtlantaFX theme	Java requirement; a desktop client also keeps the data on the student's computer.	GUI pages call core services in-process (no REST hop); REST remains only for the Chrome extension (F04).
Typer CLI	picocli CLI (`bin/careerpilot`)	Java requirement.	Same commands, one subcommand class per feature group.
PostgreSQL	H2 by default, any JDBC URL through DATABASE_URL	Zero-setup for graders; PostgreSQL still works.	Entities use JPA with generated UUIDs.
MinIO/S3 object storage and signed URLs	Local StorageService under ~/.careerpilot	No cloud dependency for a single-user tool.	Signed URLs became direct service calls (DocumentService.file, PackageAssembler.download).
ProfileFact subclass per kind	One ProfileFact entity with a FactType enum and validated JSON	Less code and a single table, validation still per type (FactValidator).	F01, F02, F10, F17 read facts through the same class.
OpenAIProvider	OpenAiCompatibleProvider and GeminiProvider; default model is Gemini through REST	A free-tier model works for development; any OpenAI-style endpoint (Groq, OpenRouter) is supported with a base URL.	LlmProvider adapter pattern unchanged.
reportlab / python-docx	Apache PDFBox / POI in DocumentRenderer	Java equivalents.	F10-F12 output the same formats (PDF, DOCX, TXT).
Gmail adapter planned as Tier 2	GmailAdapter, GmailConnector (browser consent on a loopback port, read-only scope), CredentialVault (AES-GCM)	Implemented in Period 3. Refresh tokens are encrypted at rest and no password is ever typed into CareerPilot.	F14 gains a third provider. Tested against a local stand-in for Google; not yet exercised against a live Google account.
CredentialVault.encrypt() in the Stage 1 class list	CredentialVault.put/get with AES-256-GCM, key file with owner-only permissions	Same responsibility, simpler API.	Used only by GmailAdapter.
First JavaFX version with plain tables	Redesign branch: tokens, sidebar groups, command palette, pipeline board, bearing ring	Usability audit with before and after screenshots in docs/gui-redesign.	No feature change; see docs/gui-redesign/CHANGES.md.
New: Docker image for the REST server	Dockerfile and docker-compose.server.yml	Lets the extension talk to a server on another machine.	Not a graded feature.

No pattern from Stage 1 was dropped or replaced, so no pattern needed a replacement justification.

5. AI tools and collaboration log

Tool or platform	Model	Used for	Tokens and cost
Claude Code (CLI)	Claude Sonnet 5.5	Implementation of core services, CLI, GUI, tests, UMLet generation, this report, final audit	113.7M tokens in 442 calls (0.9K uncached input, 0.75M output, 2.30M cache writes, 110.7M cache reads); about USD 24.29
Claude Code (CLI)	Claude Opus 5.5	Design and harder implementation work in earlier sessions	34.3M tokens in 171 calls (0.4K uncached input, 0.31M output, 1.41M cache writes, 32.6M cache reads); about USD 19.70
Gemini API (runtime, not development)	Gemini default model	The product's own LLM calls during early end-to-end checks	12 calls logged with 0 tokens (the free-tier quota ran out and the model answered HTTP 503); no charge
OpenAI-compatible API (runtime, not development)	gpt-6-luna	The product's own LLM calls during the Nov 3 to Nov 6 live end-to-end checks	14 calls, 21,558 input and 9,231 output tokens (30,789 total); cost not computed because no price list for this model was available, see the billing page of that account

How the numbers were obtained. The Claude Code rows are summed from the local Claude Code session transcripts for this project (every distinct API response, duplicates removed) and priced at the list rates in the local cost tracker: Sonnet 5.5 USD 2 / 10 / 2.50 / 0.10 and Opus 5.5 USD 4 / 20 / 5 / 0.20 per million input / output / cache-write / cache-read tokens. Total development cost is about USD 43.99 (147.9M tokens). These are estimates from list prices, not invoice figures, and they cover only the transcripts still stored on the author's machine. The runtime rows are read from the `LlmUsageRecord` table of the application's own database (`careerpilot llm-check` shows the same records).

Representative interactions.

1. *Resume import.* Prompt: implement F02 with ResumeParser and a schema-validated extraction. Result: ResumeParser plus the `resume.extract_facts` prompt. Reviewed by hand; extraction output is saved as unverified so a wrong guess cannot reach a document.
2. *Fabrication guard.* Prompt: generated documents must not state anything the student has not verified. Result: FabricationGuard.check on every claim and a final `generate()` that cannot skip it. Verified with tests that insert an invented employer.
3. *Gmail.* Prompt: add a Gmail provider without storing a password. Result: loopback OAuth, read-only scope, AES-GCM vault. Reviewed: the test double caught a status bug (an expired grant must mark the account for reconnect) and an old test that assumed GMAIL was unavailable.
4. *Class diagrams.* Prompt: update the UML to the implemented classes. Result: a generator that reads signatures from the code. A label calling PlaybookLibrary a Strategy was corrected because it is a library of fallback plans.

6. Development and verification summary

6.1 Weekly commit summary

Only commits that change code, tests or build files count as code commits; documentation and screenshot-only commits are listed but not counted.

Period 1 (Oct 20-26): 12 code commits (19 commits in total)

Commit	Date	Message	Changes code
fc0fbeb	2026-10-23	Backend skeleton: Django project, token auth, tenant-scoped repository, task queue	yes
972ef6e	2026-10-24	LLM subsystem: LLMClient facade with Gemini, OpenAI-compatible and mock provider adapters	yes
68e0cf2	2026-10-24	F01/F03: candidate profile, verified facts and career goal with strategy weights	no (docs/assets)
1abc24e	2026-10-24	F02: resume import with PDF/DOCX text extraction and AI fact extraction	yes
7cc81fd	2026-10-24	F04-F06: job import pipeline with source adapters, normalization and duplicate detection	yes
87fd4f1	2026-10-24	F07/F08: AI job analysis and strategy-based candidate-job matching	yes
73b539b	2026-10-24	F13: application lifecycle with State pattern, timeline, undo and status events	no (docs/assets)
25ea97e	2026-10-25	F10: tailored resume generation with FabricationGuard and Template Method pipeline	yes
6ed218b	2026-10-25	F11: tailored cover letter generation on the shared document template	yes
8f79ec0	2026-10-25	F12: application package builder with ZIP export	no (docs/assets)
26f14d8	2026-10-25	Agent tools: Command-pattern tool catalog, ToolManager invoker, approval policy and memory	no (docs/assets)
7522094	2026-10-25	F16: goal-driven agent with planning, approvals, reasoning, replanning and memory	yes
b4f4e38	2026-10-25	F09: automatic agent batch triage with budget-aware deep analysis and shortlist	no (docs/assets)

Commit	Date	Message	Changes code
97e885f	2026-10-25	GUI: React + Vite app shell, sign-in and Profile page (F01-F03)	yes
84731d7	2026-10-25	GUI: Jobs and Job detail pages (F04-F09 views)	yes
738329c	2026-10-25	GUI: application tracker with lifecycle actions, documents and package download (F10-F13)	yes
198866f	2026-10-25	GUI: agent workspace and dashboard (F09, F16)	yes
ded6e56	2026-10-25	README: local setup, key configuration, backend/GUI launch and test commands	no (docs/assets)
68b32aa	2026-10-25	Fix llm_check registration and add backoff for overloaded LLM providers	no (docs/assets)

Period 2 (Oct 27-Nov 2): 37 code commits (38 commits in total)

Commit	Date	Message	Changes code
9873d25	2026-10-28	F14: email data model, provider adapters (.eml upload, mock inbox) and normalizer	yes
7b569a0	2026-10-28	F14: rule/LLM/hybrid classification strategies, application matcher, email sync service and review API	yes
d6e9b56	2026-10-30	Java stack: Maven multi-module build, Spring Boot core, shared infrastructure and local accounts	yes
31ded35	2026-10-30	LLM subsystem in Java: LlmClient facade, Gemini/OpenAI-compatible/mock provider adapters, prompt registry, schema validation and usage budgets	yes
7531d13	2026-10-30	F01-F03: candidate profile with validated facts, resume import with AI fact extraction and career goal strategies in Java	yes
237d966	2026-10-30	F04-F06: job import pipeline in Java with source adapters, normalization and duplicate detection	yes
7bb58f3	2026-10-30	F07/F08: AI job analysis with keyword fallback and strategy-based candidate-job matching in Java	yes
58fcfc8	2026-10-30	F13: application lifecycle with State pattern, timeline, undo and optimistic version check in Java	yes

Commit	Date	Message	Changes code
f8d2eeb	2026-10-30	F10/F11: tailored resume and cover letter on a Template Method pipeline with the FabricationGuard	yes
868e70d	2026-10-30	F12: application package builder with ZIP export and checklist in Java	yes
adf5c0e	2026-10-30	Fix entity identity: generate UUIDs in Hibernate so save() persists the same instance	yes
e8a7404	2026-10-30	Agent tools: Command-pattern catalog, ToolManager invoker, approval policy and long-term memory in Java	yes
945fac7	2026-10-30	F16: goal-driven agent loop with LLM planning, playbook fallback, approvals, reasoning, replanning and memory	yes
3ce228d	2026-10-30	F09: automatic batch triage triggered by the import event, with budget-aware deep analysis and shortlist	yes
a1deda9	2026-10-30	F14: email data model, provider adapters (.eml upload, mock inbox) and normalizer in Java	yes
6867bac	2026-10-30	F14: rule, LLM and hybrid classification strategies, application matcher, email sync service and review flow	yes
987b1d3	2026-10-30	REST server for the browser extension: sign-in, token auth, batch import and job listing	yes
3b0baaa	2026-10-30	F04: Chrome extension that imports the postings on the current page into CareerPilot on click	yes
0f1ab61	2026-10-30	CLI: picocli commands for accounts, profile, jobs, applications, documents, agent and email	yes
8856b6d	2026-10-30	CLI tests and launch scripts: end-to-end runs of the real command line against the core and mock LLM	yes
2be2457	2026-10-30	JavaFX GUI: sign-in, dashboard, profile and goal, jobs, tracker, agent workspace, email review and settings	yes
6f4796a	2026-10-30	GUI render test, launch script and shared per-user data directory for GUI, CLI and server	yes

Commit	Date	Message	Changes code
e2b3e29	2026-10-30	Remove the Python/Django backend, React frontend and CLI stub; switch CI and config to the Java build	yes
3e3db4c	2026-10-30	README: Java build, local configuration, GUI/CLI/server launch commands, example usage and new layout	no (docs/assets)
334a9a2	2026-10-30	Fix: blank .env values no longer switch to an in-memory database; agent failures no longer roll back the import	yes
1ee8ee4	2026-10-30	LLM provider errors: one readable line instead of raw JSON, and a quiet CLI terminal	yes
dcb4da4	2026-10-30	Server config: use the shared DataSourceConfig instead of a duplicate datasource block	yes
3d53e76	2026-10-30	Extension: read schema.org data, job cards, header-less tables and plain posting pages, and say how the page was read	yes
7aa4d14	2026-10-30	Keep a link back to every imported posting	yes
b01f02d	2026-10-31	GUI baseline before the redesign: Icons, Theme, Toast, Interview and Insights views, plus current styles	yes
982879e	2026-10-31	GUI audit: per-page findings with HIG citations, before screenshots in light and dark, render test captures both themes	yes
e632be5	2026-10-31	GUI tokens: primitive, semantic and component layers in tokens.css and components.css, system-following light and dark theme, direction write-up	yes
3ef2a0b	2026-10-31	App shell: grouped collapsible sidebar, per-page top bar with progress, menu bar, Cmd+number navigation, command palette, branded sign-in with inline validation, toasts instead of error alerts	yes
3b48c52	2026-10-31	Dashboard: context tiles, Needs you queue as keyboard-reachable rows, shortlist with the bearing ring, empty and loading states	yes

Commit	Date	Message	Changes code
735e592	2026-10-31	Jobs: bearing ring in the list, two-line role cells, detail pane with actions on top, search and import in the top bar, empty states	yes
bbc47ae	2026-10-31	Tracker: pipeline board by state, search in the top bar, action, timeline, documents and notes panel, empty states	yes
d239791	2026-10-31	Agent: approval card opens by itself, run status and plan; Settings: appearance control, switch, grouped cards; Insights: bar charts with written values, pipeline view, empty state	yes
7115b3a	2026-10-31	Final pass: Email and Agent columns, remove legacy styles, after screenshots, CHANGES write-up	yes

Period 3 (Nov 3-9): 14 code commits (19 commits in total)

Commit	Date	Message	Changes code
f29a262	2026-11-03	F15: event-driven updates with Observer listeners, notifications and deadline reminders	yes
8b84270	2026-11-03	F17: interview preparation plan with fabrication-checked STAR stories, deterministic fallback and automatic trigger	yes
79abc86	2026-11-03	F18: AI mock interview with rubric scoring, resumable sessions and feedback memory	yes
c1e8786	2026-11-03	F19: outcome analytics and adaptive strategy with bounded, evidence-cited proposals the student accepts, rejects or reverts	yes
d9e0d68	2026-11-03	CLI: notifications and reminders, interview preparation, mock interview and analytics commands (F15, F17-F19)	yes
2a9fa27	2026-11-03	Agent: prepare_interview tool and PREPARE_INTERVIEW playbook so a goal like ‘get me ready for the Acme interview’ works	yes
00893ef	2026-11-03	README: all 19 features, new CLI commands and GUI pages	no (docs/assets)
b5fb31a	2026-11-03	CLI: show a dash instead of null when no application has a response time yet	yes

Commit	Date	Message	Changes code
0dc99e6	2026-11-03	Analytics: fallback narrative only claims ‘no change justified’ once there are 10 outcomes	yes
c154ca2	2026-11-03	Docker: image and compose file to run the REST server for the Chrome extension	yes
b4c8a0e	2026-11-03	F14: Gmail provider with read-only OAuth consent flow, AES-GCM credential vault for refresh tokens and a local-Google test double	yes
6687cc7	2026-11-03	CLI: email connect-gmail signs in through the browser and syncs the mailbox	yes
242cc6c	2026-11-03	GUI: Connect Gmail button on the Email page opens the consent screen in the browser and syncs	yes
5beb6cc	2026-11-03	Stage 2 UMLet class diagrams generated from the implemented Java classes (agent, state/observer, strategy/adapter, LLM/templates, period 3 features)	no (docs/assets)
0e51ded	2026-11-03	Stage 2 report generator with source-checked feature mapping, patterns, changes, commit tables, demo script and Gmail setup in the README	no (docs/assets)
e07041e	2026-11-03	Stage 2 report: refresh commit tables	no (docs/assets)
9cdf40c	2026-11-03	CI: quote the build step name; the unquoted colon made the workflow file invalid so no job ever started	yes
53e8abb	2026-11-03	LLM: send max_completion_tokens and no temperature to the official OpenAI endpoint, whose current models reject max_tokens (found with a live key)	yes
fa75259	2026-11-03	Stage 2 report: record the live end-to-end verification, the OpenAI parameter defect and current test counts	no (docs/assets)

6.2 Checks performed

- `mvn test` runs all modules: core 124 tests (services, state machine, agent loop with approval and undo, fabrication guard, email pipeline, interview, analytics, Gmail flow), server 3, CLI 7, GUI render test.
- Extension unit tests with node (`extension/tests`).

- Gmail flow tested end to end against a local HTTP server that plays Google: consent redirect with state check, code exchange, encrypted token on disk, message fetch, revoked grant, cancelled consent, missing configuration.
- Earlier manual run with the real Gemini key: resume extraction, triage, agent run and email classification.
- Live end-to-end run on Nov 3 with a real OpenAI-compatible model (isolated scratch database): resume extraction (12 facts), goal and strategy, CSV import with the bad row reported, automatic triage, analysis and match (92/100 for the best job), resume and cover letter (fabrication guard PASSED), package ZIP, lifecycle transitions with notifications, interview preparation plan, mock interview with rubric scores and summary, agent goal “prepare my interview” (2 steps, 2152 tokens), analytics (correctly refuses to propose changes with 0 outcomes), demo inbox email sync (3 emails, 1 not job related, 1 applied automatically, 1 for review). About 16,000 tokens in total.

6.3 Important defects found and corrected

- Preassigned UUIDs made `save()` merge and detach instances; entities now use generated UUIDs.
- A failing agent tool poisoned the surrounding import transaction; `TransactionConfig` now keeps rollback local to the failed participant.
- Blank `DATABASE_URL=` lines in an old `.env` were treated as a value; blank now means unset.
- Analytics fallback text claimed “your data does not justify a change” with too few outcomes; it now says so only from 10 outcomes.
- CLI printed `null` for the median response time when there was no response yet.
- Found with the live key: current OpenAI models reject `max_tokens` and a custom temperature; the provider now sends `max_completion_tokens` for the official endpoint (unit-tested).

6.4 Remaining limitations

See Section 1. The unit tests use the mock LLM, so wording quality of live answers is not covered by tests.

7. Demonstration index

Each feature needs a labelled video segment showing the input, the running application, the output, where AI is involved and one error case. The full demonstration is one recording: [CareerPilot demo video](). It shows the GUI and the CLI running, and each feature below links to it.

Feature	How it is invoked	Video
F01 Candidate profile and verified facts	GUI Profile page; CLI `profile show/set/fact-*`	https://drive.google.com/file/d/1NThfqULr6I2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F02 Resume import and AI fact extraction	GUI Profile > Import resume; CLI `profile resume FILE`	https://drive.google.com/file/d/1NThfqULr6I2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F03 Career goal and strategy configuration	GUI Profile > Goal; CLI `profile goal-set/goal-show`	https://drive.google.com/file/d/1NThfqULr6I2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F04 Browser-extension batch job import	Chrome extension (extension/content, popup, api); REST `/api/import`; CLI `jobs import-york`	https://drive.google.com/file/d/1NThfqULr6I2PLUaawXwYJkWKZ8aottcT/view?usp=sharing

Feature	How it is invoked	Video
F05 Manual job import (paste or file)	GUI Jobs > Import; CLI `jobs import-paste/import-file`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F06 Job normalization and duplicate detection	Runs inside every import	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F07 AI job description analysis	GUI Jobs detail; CLI `jobs analyze`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F08 Candidate-job matching and skill-gap analysis	GUI Jobs (bearing ring, detail); CLI `jobs match/score/show`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F09 Agent batch triage and shortlist	Automatic after an import; GUI Agent page; CLI `agent runs/show`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F10 Tailored resume with fabrication guard	GUI Tracker > Documents; CLI `docs resume`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F11 Tailored cover letter	GUI Tracker > Documents; CLI `docs cover-letter`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F12 Application package builder and ZIP export	GUI Tracker > Build package; CLI `docs package/download`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F13 Application lifecycle tracking and undo	GUI Tracker board; CLI `apps create/act/undo`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F14 Email monitoring and application event detection	GUI Email page; CLI `email upload/sync-mock/connect-gmail/events/confirm`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F15 Event-driven updates, reminders and notifications	GUI Dashboard feed; CLI `notifications`, `remind`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F16 Goal-driven agent workspace	GUI Agent page; CLI `agent run/approve/reject/undo/memory`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F17 Interview preparation plan	GUI Interview page; CLI `interview prepare/show`; agent tool prepare_interview	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F18 AI mock interview and feedback	GUI Interview page; CLI `interview mock/answer/finish`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing
F19 Outcome analytics and adaptive strategy	GUI Insights page; CLI `analytics`	https://drive.google.com/file/d/1NThfqULr6l2PLUaawXwYJkWKZ8aottcT/view?usp=sharing